the red colour is determined by the number of R genes in the F_2 generation.

Four R genes: A dark red kernel colour is obtained.**Three R genes:** Medium - dark red kernel colour is obtained.**Two R genes:** Medium-red kernel colour is obtained.**One R gene:** Light red kernel colour is obtained.**Absence of R gene:**Results in White kernel colour.

The R gene in an additive manner produces the red kernel colour. The number of each phenotype is plotted against the intensity of red kernel colour which produces a bell shaped curve. This represents the distribution of phenotype. Other example: Height and skin colour in humans are controlled by three pairs of genes.

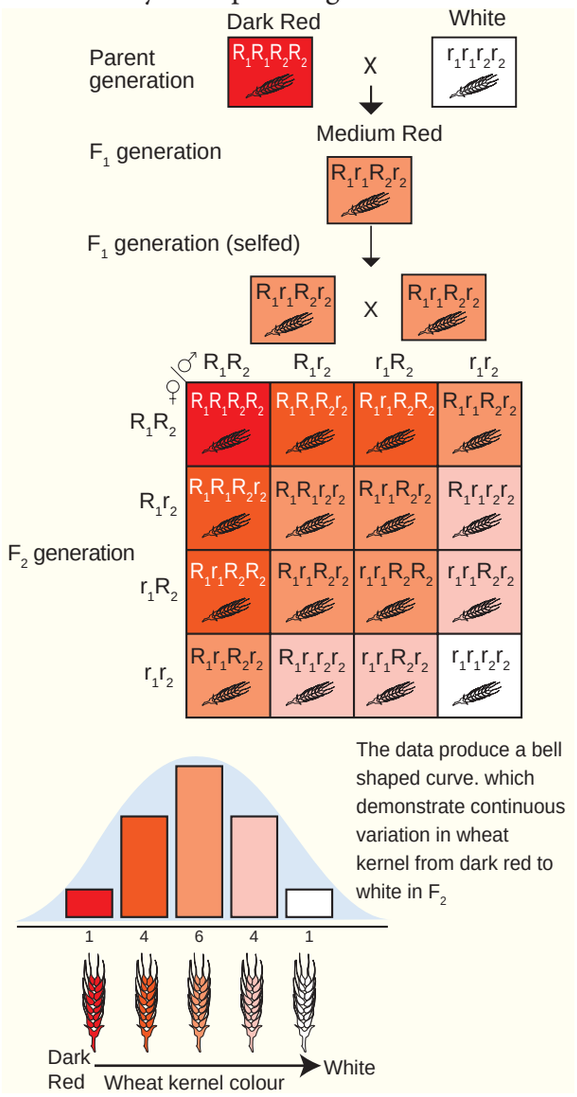


Figure 2.17 (a): Polygenic inheritance in wheat kernel colour

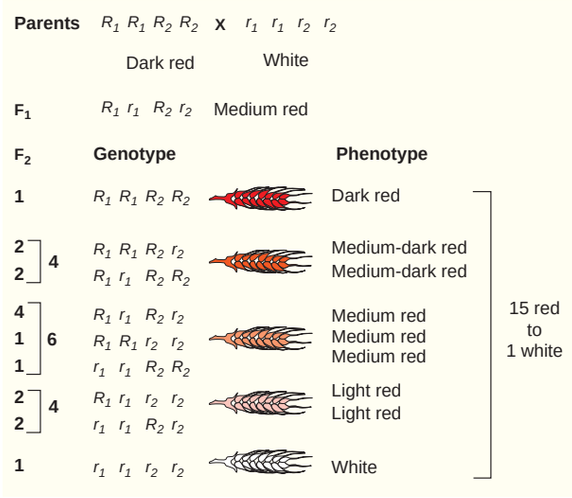


Figure 2.17 (b) : The genetic control of colour in wheat kernels.

Conclusion:

Finally the loci that was studied by Nilsson – Ehle were not linked and the genes assorted independently.

Later, researchers discovered the third gene that also affect the kernel colour of wheat. The three independent pairs of alleles were involved in wheat kernel colour. Nilsson – Ehle found the ratio of 63 red : 1 white in F_2 generation – 1 : 6 : 15 : 20 : 15 : 6 : 1 in F_2 generation.

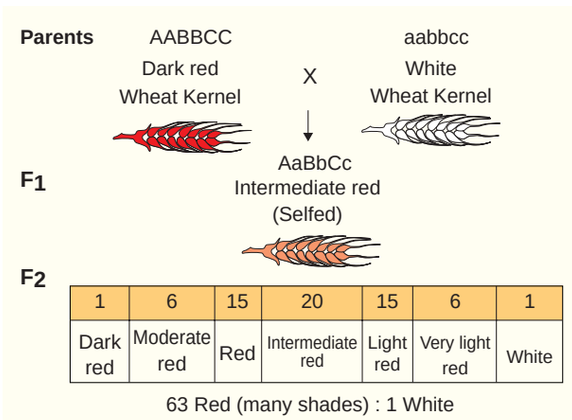


Figure 2.18: Polygenic inheritance in Wheat kernel

From the above results Nilsson – Ehle showed that the blending inheritance was not taking place in the kernel of wheat. In F_2 generation plants have kernels with wide range

TN_GOVT_BIOLOGY-BOTANY_XII_PAGES_033-052 CH 02.indd 44 3/14/2020 10:30:54 AM